

ch57xGUI - Project Handoff Summary

Date: Thursday, 22 January 2026 (Australia/Sydney)

Status: Working baseline restored ("the very first code")

Overview

ch57xGUI is a personal, minimal GUI wrapper for **ch57x-keyboard-tool** (CH57x macro pad programmer). It provides a desktop-friendly way to edit the YAML configuration: select a button/knob, type or build a mapping, save/load YAML, validate, and upload via the CLI tool.

Key goals met

- Create a usable GUI that avoids terminal-only workflows for everyday remapping.
- Expose the YAML schema (layers, buttons grid, knobs) with a simple selection + editor flow.
- Support raw mapping strings and multi-step sequences (comma separated, up to 5 chords).
- Integrate CLI actions: validate and upload using ch57x-keyboard-tool on PATH.

Repo and main files

Repository: GitHub: k0de14242/ch57xGUI

Primary file: main.py (single-file app)

How to run (local dev)

Assuming the project lives at ~/Desktop/ch57xGUI:

```
cd ~/Desktop/ch57xGUI
python3 -m venv .venv
source .venv/bin/activate
pip install -U pip
pip install pyside6 pyyaml
python3 main.py
```

Tool dependency (CLI)

The GUI shells out to **ch57x-keyboard-tool** for *show-keys*, *validate*, and *upload*. It must be available on PATH as **ch57x-keyboard-tool**. If the GUI log says it is not found, verify in Terminal:

```
ch57x-keyboard-tool --help
```

User workflow

- **Config:** New / Load / Save / Save As.
- **Keyboard metadata:** orientation, rows, columns, knobs. Changing these rebuilds the UI and reshapes layer arrays.
- **Selection:** click a button (row,col) or knob action (CCW/PRESS/CW).
- **Edit mapping:** use Mapping String (raw) or Sequence list (comma-separated steps).
- **Recording:** Record chord or Record sequence (Enter=done, Esc=cancel).
- **Builder:** dropdown key + modifier checkboxes; can set raw or append a step; also supports modifiers-only step.
- **CLI:** Validate and Upload use the current in-memory YAML via stdin.

Data model

The app maintains a Python dict matching the YAML schema. Defaults are embedded in **DEFAULT_CONFIG** and loaded YAML is normalized by **ensure_config_shape** to avoid missing keys or bad shapes.

Key mapping format

- A mapping is either:
- a key chord string (e.g., **ctrl-v, alt, win-rctrl-backspace**)
 - a sequence of up to 5 chords separated by commas (e.g., **ctrl-v,ctrl-c**)
 - a mouse event string (e.g., **click+rclick, shift-wheelup**)
 - a raw scan code token like **<101>**

Known limitations and current stance

- **macOS key capture:** the current baseline capture path uses Qt's *event.text()* for letters/punctuation. On macOS, Option can produce composed characters (e.g., Option+A -> å), and Cmd/Ctrl combos may produce empty text. This makes capture imperfect for those chords.
- **Practical workaround:** use the **Builder** controls to set chords precisely (modifiers + key) without relying on live capture. Since you already know which keys are which, this is an acceptable trade-off and keeps the app working.
- **Cosmetic labels:** baseline uses tool tokens (alt/win/enter). Any display-friendly label work (option/cmd/return) is intentionally deferred because it became a time sink and did not improve core functionality.

Troubleshooting

- **GUI opens but validate/upload fails:** check PATH for ch57x-keyboard-tool; run it from Terminal to confirm.
- **YAML loads but UI looks wrong:** verify rows/columns/knobs match the physical device model; then reload.
- **Nothing records during Record chord/sequence:** click the capture field (read-only but still focusable) and try again; use Builder if macOS composition interferes.

Next steps (if you ever feel like it)

- **Robust chord capture:** implement physical key capture using `ev.key()` + `ev.modifiers()` with a layout-aware mapping for punctuation; treat Shift-only separately; keep a fallback for raw tokens.
- **Better knob selection highlighting:** currently button highlighting is solid; knob action highlighting is best-effort.
- **Packaging:** optional - create a macOS app wrapper (e.g., `python -m PyInstaller`) so it launches from an icon without opening Terminal.
- **Config UX:** add quick actions for copy/paste mappings, duplicate layer, and per-layer rename.

Current baseline definition

This handoff corresponds to the working baseline identified by you as: "**the very first code you gave me**". From here, avoid incremental whitespace-sensitive edits in chat; prefer atomic replacement or local patch scripts.

End of handoff.